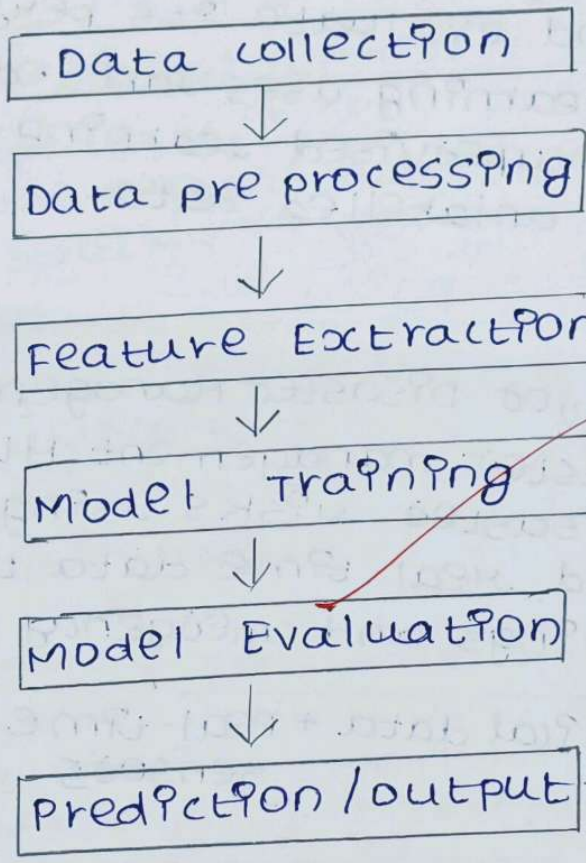
CO2-AT2

Scenario - Based Questions

Name :- Harshith G1
Reg NO :- 192525394
course :- ITA0615
Machine Learning

1) Functional Blocks:

A machine learning system learns the patterns from data and uses them to make predictions or discover useful information.



* Data collection: collects the data from sensors, databases or surveys.
* Data preprocessing: removes errors, handling missing values and convert into suitable format.
* Feature Extraction: Identifies characteristics of attributes from data.
* Model Training: uses an algorithm to learn patterns from data.

*Model Evaluation: Measures model performance using metrics such as accuracy, precision or recall

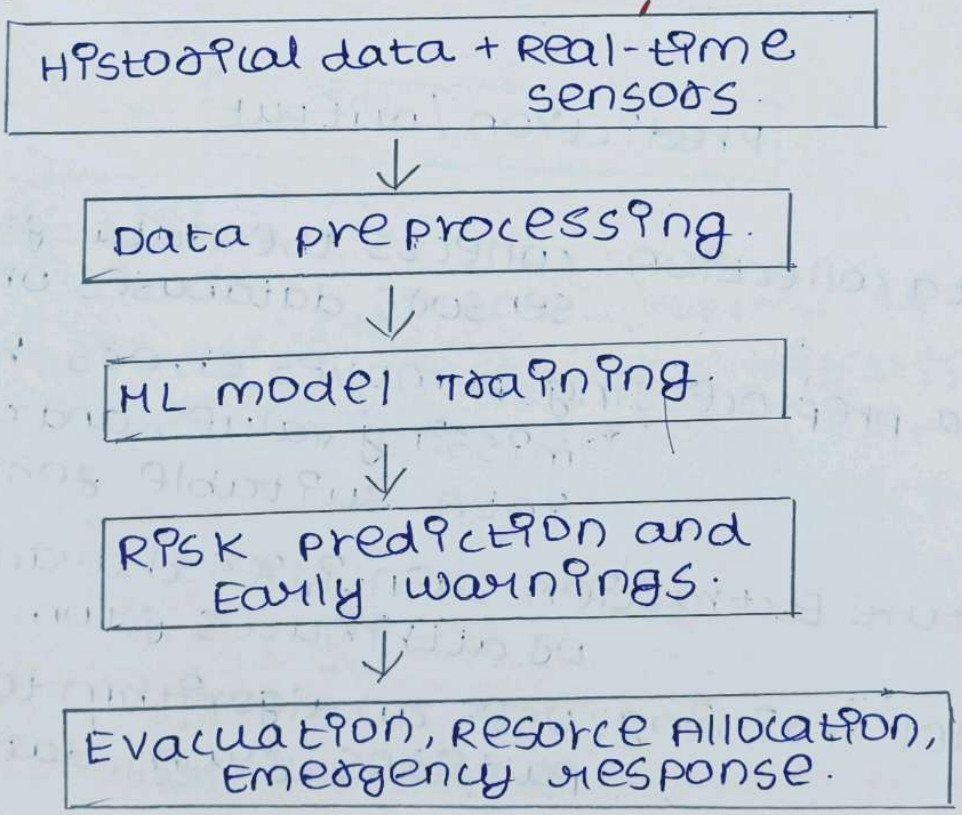
*prediction/output: uses the trained model to predict outcomes for the new data.

conclusion:-

An ML system collects data, learn the patterns and evaluate its performance. supervised learning uses labelled data, whereas unsupervised learning discovers patterns in unlabelled data.

2) ML system for Disaster Management:

A disaster management ML system predicts disaster risks using historical records and real-time data to support early warnings and emergency planning.



Data processing :-

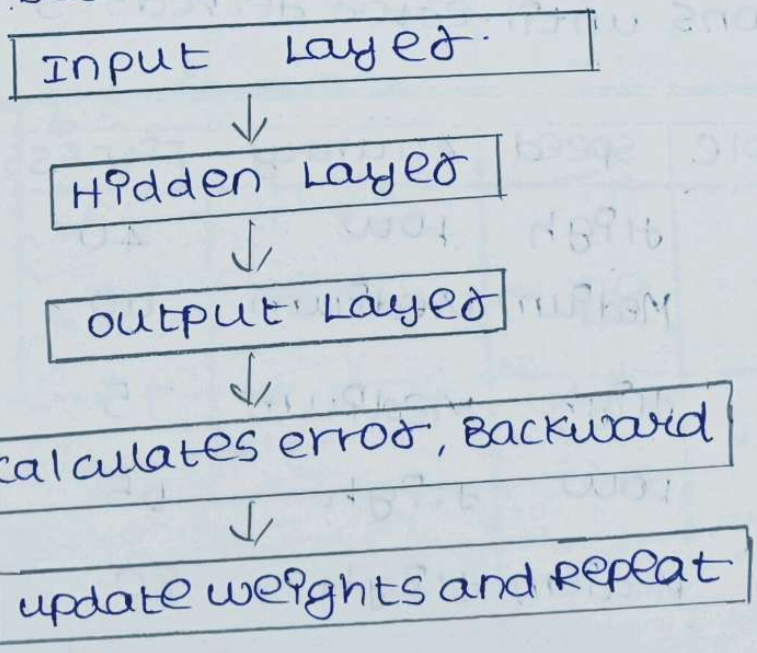
* Remove duplicate records.
* Normalize Numerical values.
* Encode categorical data.
* Select important features using correlation analysis or domain knowledge.
* Organize historical data chronologically.

Training and Evaluation :-

* Split historical data into training, validation and test sets.
* Train the selected models using historical disaster data.
* Tune parameters using validation data.
* Evaluate using precision, recall, F1 score and false alarm rate.
* Test real-time predictions and continuously monitor performance.

3) Backpropagation in Multilayer perceptron :-

It is a learning algorithm that calculates the error between actual and predicted output.



Working :-
consided a binary classification problem whe the network predicts that a student passe an examination.

1) Forward pass :- Input features are passed through the input and hidden layers to generate a prediction. suppose o/p is 0.7.

2) Error calculation :- If actual target is 1, the prediction error can be calculated by

$$E = \frac{1}{2}(y - \hat{y})^2$$

$$= \frac{1}{2}(1 - 0.7)^2$$

$$= 0.045$$

3) Backward propagation :- The error is propagated backward through output and hidden layer using chain rule.

4) Weight update :

$$W_{new} = W_{old} - \eta \frac{\partial E}{\partial w}.$$

5) Repeat : This process continues for multiple iterations until error decreases.

4)

Example	speed	Accuracy	Fitness	suitable
1	High	Low	40	NO
2	Medium	Medium	60	yes
3	High	Medium	75	yes
4	Low	High	65	yes
5	Medium	High	80	yes

Steps in Genetic Algorithm :-

1) Initial population :-
Each row represents candidate solution. Its fitness value measures its quality.

2) selection : solutions with higher fitness value are more likely to be selected as parents. Therefore Ex 5 and Ex 3 are strong candidates.

3) crossover :-
- parent 1 : (High speed, Medium Accuracy)
- parent 2 : (Medium speed, High Accuracy)
- offspring : (High speed, High Accuracy).

4) Mutation :
A feature is randomly changed to explore new solutions. For example, an offspring's speed may change from medium to High.

5) Given :-

Example	Actual	predicted	classification
1	yes	yes	TP
2	yes	NO	FN
3	NO	NO	TN
4	yes	yes	TP
5	NO	yes	FP.

- **Accuracy:**

$$= \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

$$= \frac{2+1}{5} \times 100 \quad \boxed{= 60\%}$$

- **Precision:**

$$= \frac{TP}{TP + FP} \times 100$$

$$= \frac{2}{3} \times 100 \quad \boxed{= 66.67\%}$$

- **Sensitivity:**

$$= \frac{TP}{TP + FN} \times 100$$

$$= \frac{2}{3} \times 100 \quad \boxed{= 66.67\%}$$

- **F1-score:**

$$= \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\boxed{= 66.67\%}$$

The model achieves 60% accuracy. It correctly identifies two patients with disease and one without disease. However it misses one positive case and incorrectly identifies one negative case as positive. The dataset is too small to establish real-world performance.